Convolutional Networks and Autoencoders

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PhD Application

March 12, 2026

Previous Work

- Neural Networks,

Previous Work

- Neural Networks,
  a) Inverse design with autoencoders,

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  b) Object detection with convolutional networks.

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Framing the problem

Questions used to frame the project:

- What is the end-goal?
- Can it be split into subgoals?
- Estimated time to completion?
- Are there alternative ways to solve the problem?

Generative AI

Outline and results:[1]

Completed (first 2–3 months):

- **Survey** deep learning models → selected VAE (Fig. 1a),
- **Implement** simplified model (Keras 3),
- **Deploy** quantise model ONNX model in browser,
- **Package** document, test, notebooks.

[1] Done at the University of Glasgow.

Generative AI

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Planned:

- **Optimise** UV HOMO-LUMO gap,
- **Select** candidate for robotic synthesis.

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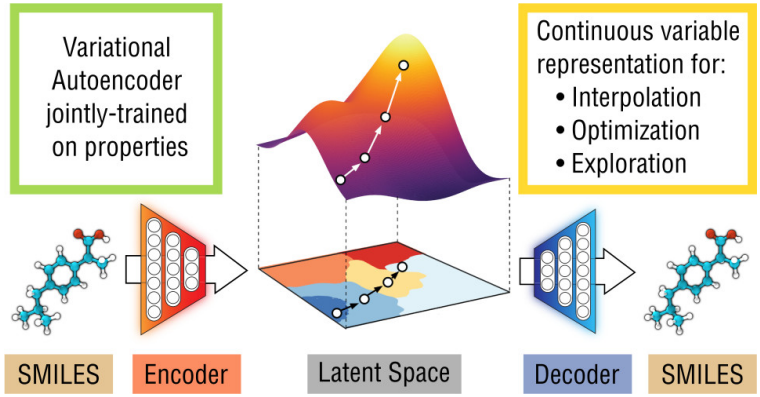
Generative AI - Inspiration



(a) VAE: Aspuru-Guzik et. al.

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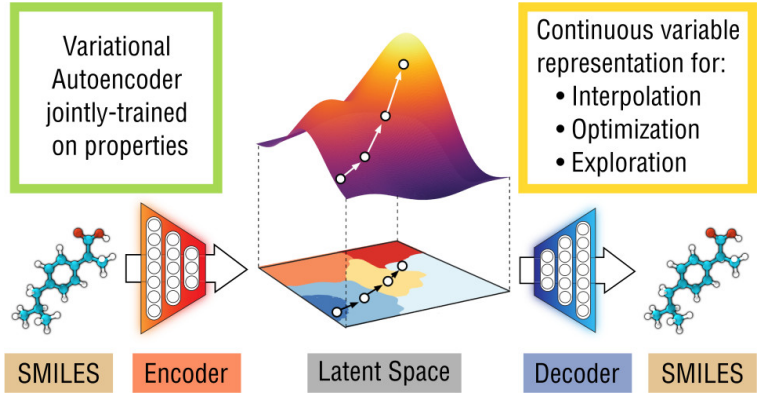

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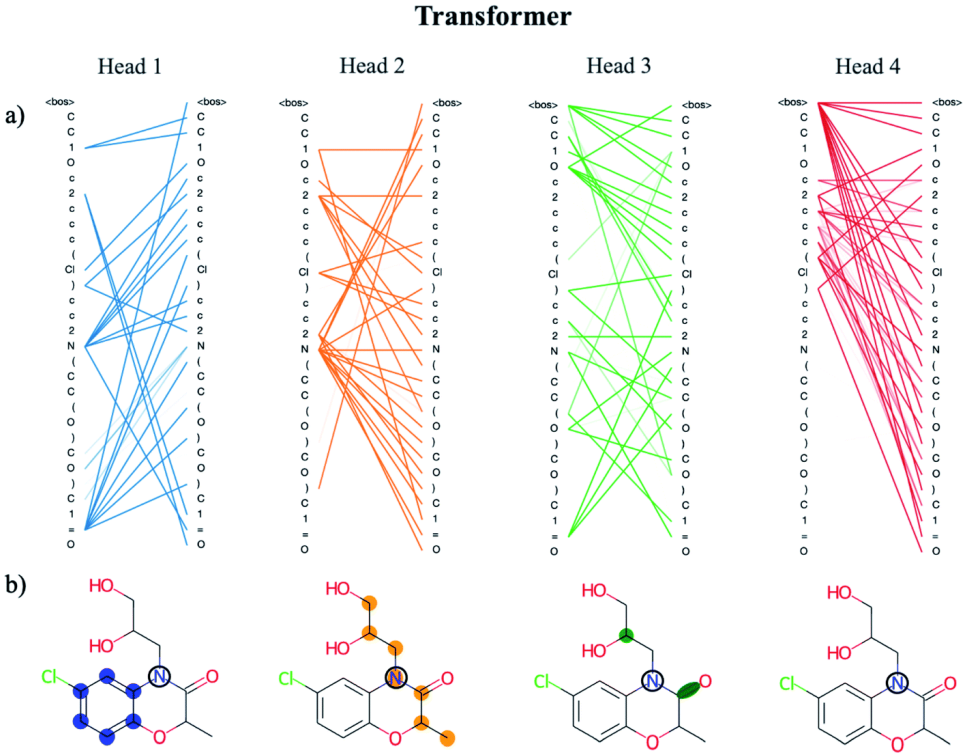
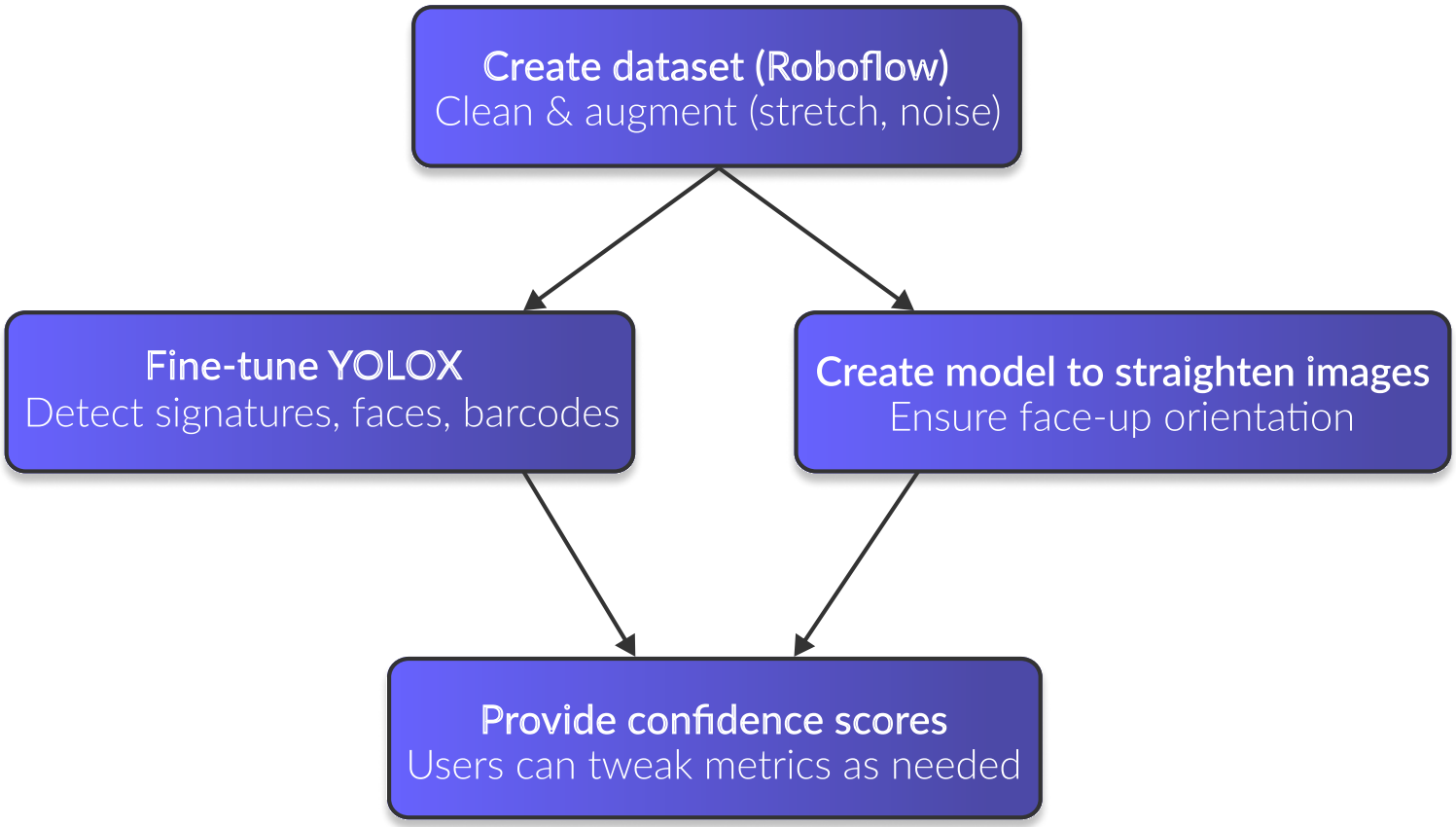


(b) Transformer Orion Dollar et. al.

Object Detection

Figure 2: Project's Development Stages



YOLOX: https://github.com/Megvii-BaseDetection/YOLOX

Object Detection

Pseudonymisation of Identity Documents at Zakodium.[2]

Frameworks: Tensorflow.js , Pytorch .

[2] Demo: https://mrz.zakodium.com/

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(a) Face and signature detection (b) Face, barcode, MRZ detection

Figure 3: © European Union, 1998–2026.

[2] Demo: https://mrz.zakodium.com/

Other Projects

Base URL for projects: https://github.com/santi-mir/⟨project⟩.

Github Project	npm / month	Description
`/spc-parser`	downloads 1.3k/month	GRAMS `.spc` parser
`/wdf-parser`	downloads 804/month	WDF Raman file parser
`/iobuffer`	downloads 19M/month	Binary data manipulation library
`/regression-simple-linear`	downloads 589k/month	Simple linear regression
`/regression-polynomial`	downloads 511k/month	Polynomial regression
`/fcnnls`	downloads 3.2k/month	Fast combinatorial non-negative least squares

Most are also published in Zenodo.

External Image Sources

- ID cards, Figs. 3a and 3b: https://commons.wikimedia.org/wiki/Category:Identity_cards_of_France
- VAE (2018) Fig. 1a: https://pubs.acs.org/doi/full/10.1021/acscentsci.7b00572
- Transformer (2021) Fig. 1b: https://pubs.rsc.org/en/content/articlehtml/2021/sc/d1sc01050f, the image was modified (cropped) to fit into the slide.

Thanks!